

# Fairness Auditing & Mitigation in Adaptive Learning Pathway Recommendation

*A real-world study on the OULAD educational dataset*

## WHAT

Test 4 recommender types on real student data

## WHY

Find which methods are unfair, and to which students

## HOW

Compare 2 fixes; report accuracy vs fairness trade-off

# Dataset, Research Questions & Hypotheses

*What we are studying and what we expect to find*

## OULAD

*Open University Learning Analytics Dataset*

32,593 real students  
22 course presentations  
~10M click records  
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### Student groups we check:

Gender  
Age band  
Disability  
Socioeconomic (IMD)

### RQ1

Of 4 recommender types (Popularity, CF, SASRec, LLM), which are most accurate AND most fair?

**What we expect:** *The deep model (SASRec) will be the best balance. The LLM will be most accurate but least fair.*

### RQ2

Of 2 fixes (fair training vs reranking output), which reduces unfairness more, and what's the accuracy cost?

**What we expect:** *Reranking will lose less than 5% accuracy — the practical fix.*

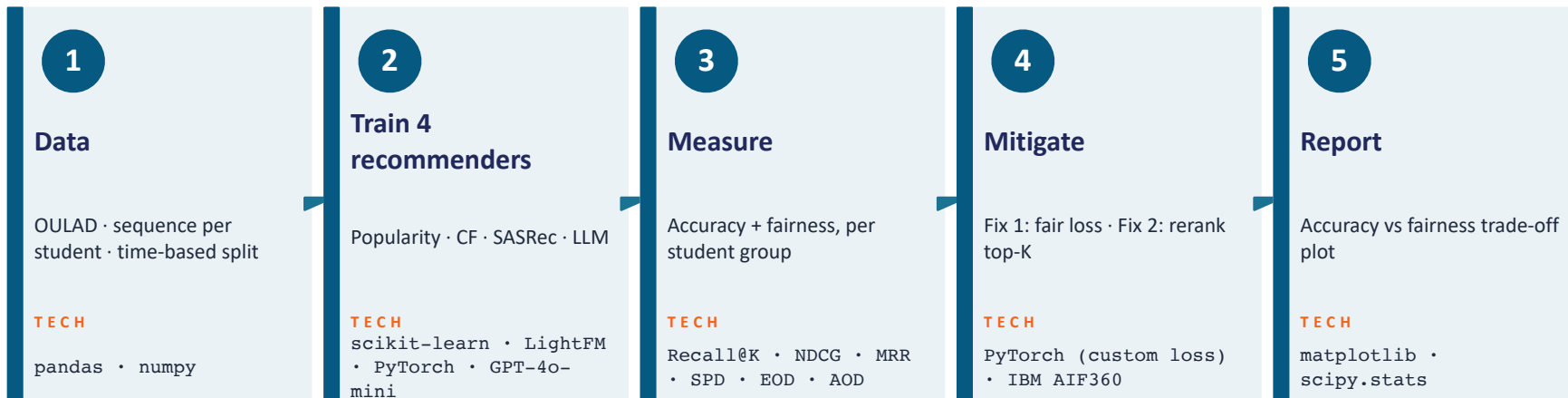
### RQ3

Is the unfairness worse for some student groups than others (gender vs IMD vs age vs disability)?

**What we expect:** *Unfairness will be largest for socioeconomic group (IMD), smallest for gender.*

# Methodology Pipeline & Tech Stack

*From raw data to a fairness-mitigated recommender*



## Full tech stack & experimental controls

**Language:** Python 3.10+ · **ML / DL:** PyTorch · scikit-learn · LightFM · **LLM:** OpenAI API (GPT-4o-mini) · **Fairness:** IBM AIF360 ·

**Stats:** scipy.stats (paired Wilcoxon, bootstrap CIs) · **Repro:** 5 seeds · git · requirements.txt · **Controls:** ceteris paribus, models tuned for accuracy only

# What We Compare & What We Measure

*Four recommender types · two fixes · three fairness metrics*

## Recommender types

| Type                    | What it does                                  | Implementation               |
|-------------------------|-----------------------------------------------|------------------------------|
| Popularity              | Recommends what most students did next        | numpy + collections (custom) |
| Collaborative filtering | Recommends what similar students did          | LightFM (WARP loss)          |
| SASRec (deep model)     | Self-attentive neural network on sequences    | PyTorch (2-layer, 50-dim)    |
| LLM-based               | Prompts a language model with student history | OpenAI API · GPT-4o-mini     |

## Metrics

### Accuracy:

Recall@10 · NDCG@10 · MRR

### Fairness (per group):

SPD · EOD · AOD · per-group Recall gap

*Computed via IBM AIF360*

## Fixes for unfairness

### Fix 1 — Train for fairness:

FairSR-style loss = accuracy +  $\lambda$  · group-divergence

### Fix 2 — Rerank output:

Post-hoc top-K rerank to balance group exposure

*Sweep  $\lambda$  to draw the trade-off curve*

# Timeline & Key References

*What we deliver week by week, and what we build on*

## Week-by-week plan

| Week      | What we achieve                                             |
|-----------|-------------------------------------------------------------|
| This week | Data pipeline, fairness module, popularity baseline running |
| Week 2    | Collaborative filtering baseline added                      |
| Week 3    | SASRec deep model trained on OULAD                          |
| Week 4    | LLM-based recommender integrated; all 4 evaluated           |
| Week 5    | Both fixes (training + reranking) implemented               |
| Week 6    | Full experiments run with 5 seeds; results collected        |
| Week 7-8  | Write the research paper (10 July)                          |
| Week 9    | Presentation prep (16-17 July)                              |

## Key references

### PRIMARY

**Li, Hsu & Zhang (2022)**

*FairSR: Fairness-aware Sequential Recommendation*  
ACM TIST 13(1)

### SUPPORTING

**Kuzilek, Hlosta & Zdrahal (2017)**

*OULAD dataset paper*

**Kang & McAuley (2018)**

*SASRec*

**Baker & Hawn (2021)**

*Algorithmic Bias in Education*